

Carnegie Mellon University

Data Analysis and Machine Learning

Polars: Reshaping Data

Introduction to Visualization

Agenda

- *More on dataframe reshaping*
- *Effective visualization*
- *Data types*
- *Mapping data to visual ques*
- *Grammar of Graphics and Plotnine*

Two Changes Next Week (Sunday, March 30)

Lecture Course-2026 Spring			
	ET Time (12 hrs time difference)	China Time	
Lecture 1	6-8AM on Saturday, March 14	6-8PM on Saturday, March 14	
Lecture 2	6-8:20AM on Saturday, March 21	6-8:20PM on Saturday, March 21	
Lecture 3	6-8:20AM on Saturday, March 28	6-8:20PM on Saturday, March 28	Homework 1
Lecture 4	6-8AM on Saturday, April 4	6-8PM on Saturday, April 4	Homework 2
Lecture 5	6-8AM on Saturday, April 11	6-8PM on Saturday, April 11	
Mid-term (Specific Time TBD, on the weekend of April 18/19)			
Lecture 6	6-8AM on Saturday, April 25	6-8PM on Saturday, April 25	
Lecture 7	6-8AM on Saturday, May 2	6-8PM on Saturday, May 2	
Lecture 8	6-8AM on Saturday, May 9	6-8PM on Saturday, May 9	Homework 3
Lecture 9	6-8AM on Saturday, May 16	6-8PM on Saturday, May 16	Homework 4
Lecture 10	6-8AM on Saturday, May 23	6-8PM on Saturday, May 23	
Final Term (Specific Time TBD, on the weekend of May 30/31)			

Takeaways from last time



1. Each **variable** forms a **column**
2. Each **observation** forms a **row**
3. Each type of **observational unit** forms a **table**

Tidy Data

$$\mathbf{F} = \mathbf{M} \mathbf{A}$$

f_1 m_1 a_1

f_2 m_2 a_2

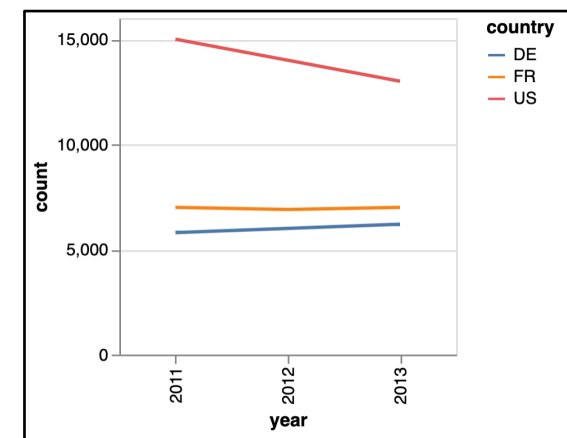
f_3 m_3 a_3

Hadley Wickham

- **Variable** - A quantity, quality, or property that you can measure
- **Value** - The state of a variable when you measure it
- **Observation** - The values of several variables measured under similar conditions.

countries				
	country	2011	2012	2013
0	FR	7000	6900	7000
1	DE	5800	6000	6200
2	US	15000	14000	13000

visualize



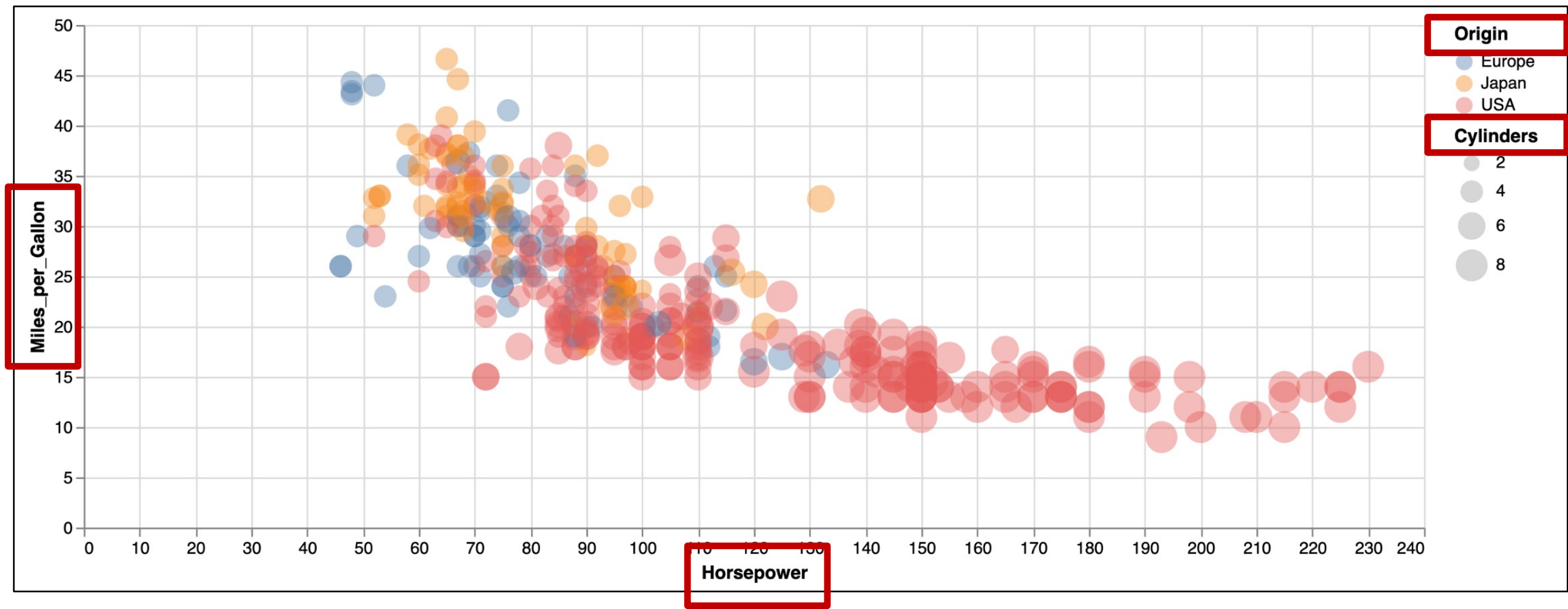
estimate

Predict:
US / 2014 ?

Two uses of Data ... **Visualization** and **Prediction**

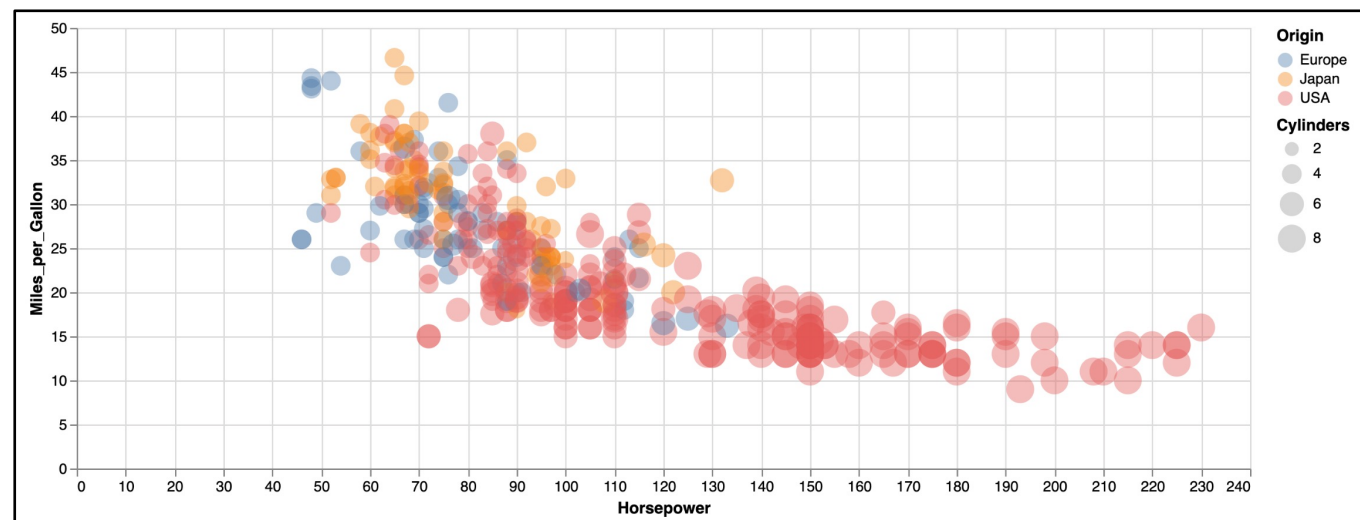
	1970 Model Cars								
	Name	Miles_per_Gallon	Cylinders	Displacement	Horsepower	Weight_in_lbs	Acceleration	Year	Origin
0	chevrolet chevelle malibu	18.0	8	307.0	130.0	3504	12.0	1970-01-01	USA
1	buick skylark 320	15.0	8	350.0	165.0	3693	11.5	1970-01-01	USA
2	plymouth satellite	18.0	8	318.0	150.0	3436	11.0	1970-01-01	USA
3	amc rebel sst	16.0	8	304.0	150.0	3433	12.0	1970-01-01	USA
4	ford torino	17.0	8	302.0	140.0	3449	10.5	1970-01-01	USA
5	ford galaxie 500	15.0	8	429.0	198.0	4341	10.0	1970-01-01	USA
6	chevrolet impala	14.0	8	454.0	220.0	4354	9.0	1970-01-01	USA
7	plymouth fury iii	14.0	8	440.0	215.0	4312	8.5	1970-01-01	USA
8	pontiac catalina	14.0	8	455.0	225.0	4425	10.0	1970-01-01	USA
9	amc ambassador dpl	15.0	8	390.0	190.0	3850	8.5	1970-01-01	USA
10	citroen ds-21 pallas	NaN	4	133.0	115.0	3090	17.5	1970-01-01	Europe
11	chevrolet chevelle concours (sw)	NaN	8	350.0	165.0	4142	11.5	1970-01-01	USA
12	ford torino (sw)	NaN	8	351.0	153.0	4034	11.0	1970-01-01	USA
13	plymouth satellite (sw)	NaN	8	383.0	175.0	4166	10.5	1970-01-01	USA
14	amc rebel sst (sw)	NaN	8	360.0	175.0	3850	11.0	1970-01-01	USA
15	dodge challenger se	15.0	8	383.0	170.0	3563	10.0	1970-01-01	USA
16	plymouth 'cuda 340	14.0	8	340.0	160.0	3609	8.0	1970-01-01	USA
17	ford mustang boss 302	NaN	8	302.0	140.0	3353	8.0	1970-01-01	USA
18	chevrolet monte carlo	15.0	8	400.0	150.0	3761	9.5	1970-01-01	USA
19	buick estate wagon (sw)	14.0	8	455.0	225.0	3086	10.0	1970-01-01	USA

Map "data" to "visual elements"



What are we trying to do in a visualization? Altair

	Name	Miles_per_Gallon	Cylinders	Displacement	Horsepower	Weight_in_lbs	Acceleration	Year	Origin
0	chevrolet chevelle malibu	18.0	8	307.0	130.0	3504	12.0	1970-01-01	USA
1	buick skylark 320	15.0	8	350.0	165.0	3693	11.5	1970-01-01	USA
2	plymouth satellite	18.0	8	318.0	150.0	3436	11.0	1970-01-01	USA
3	amc rebel sst	16.0	8	304.0	150.0	3433	12.0	1970-01-01	USA
4	ford torino	17.0	8	302.0	140.0	3449	10.5	1970-01-01	USA
5	ford galaxie 500	15.0	8	429.0	198.0	4341	10.0	1970-01-01	USA
6	chevrolet impala	14.0	8	454.0	220.0	4354	9.0	1970-01-01	USA
7	plymouth fury iii	14.0	8	440.0	215.0	4312	8.5	1970-01-01	USA
8	pontiac catalina	14.0	8	455.0	225.0	4425	10.0	1970-01-01	USA
9	amc ambassador dpl	15.0	8	390.0	190.0	3850	8.5	1970-01-01	USA
10	citroen ds-21 pallas	NaN	4	133.0	115.0	3090	17.5	1970-01-01	Europe
11	chevrolet chevelle concours (sw)	NaN	8	350.0	165.0	4142	11.5	1970-01-01	USA
12	ford torino (sw)	NaN	8	351.0	153.0	4034	11.0	1970-01-01	USA
13	plymouth satellite (sw)	NaN	8	383.0	175.0	4166	10.5	1970-01-01	USA
14	amc rebel sst (sw)	NaN	8	360.0	175.0	3850	11.0	1970-01-01	USA
15	dodge challenger se	15.0	8	383.0	170.0	3563	10.0	1970-01-01	USA
16	plymouth 'cuda 340	14.0	8	340.0	160.0	3609	8.0	1970-01-01	USA
17	ford mustang boss 302	NaN	8	302.0	140.0	3353	8.0	1970-01-01	USA
18	chevrolet monte carlo	15.0	8	400.0	150.0	3761	9.5	1970-01-01	USA
19	buick estate wagon (sw)	14.0	8	455.0	225.0	3086	10.0	1970-01-01	USA

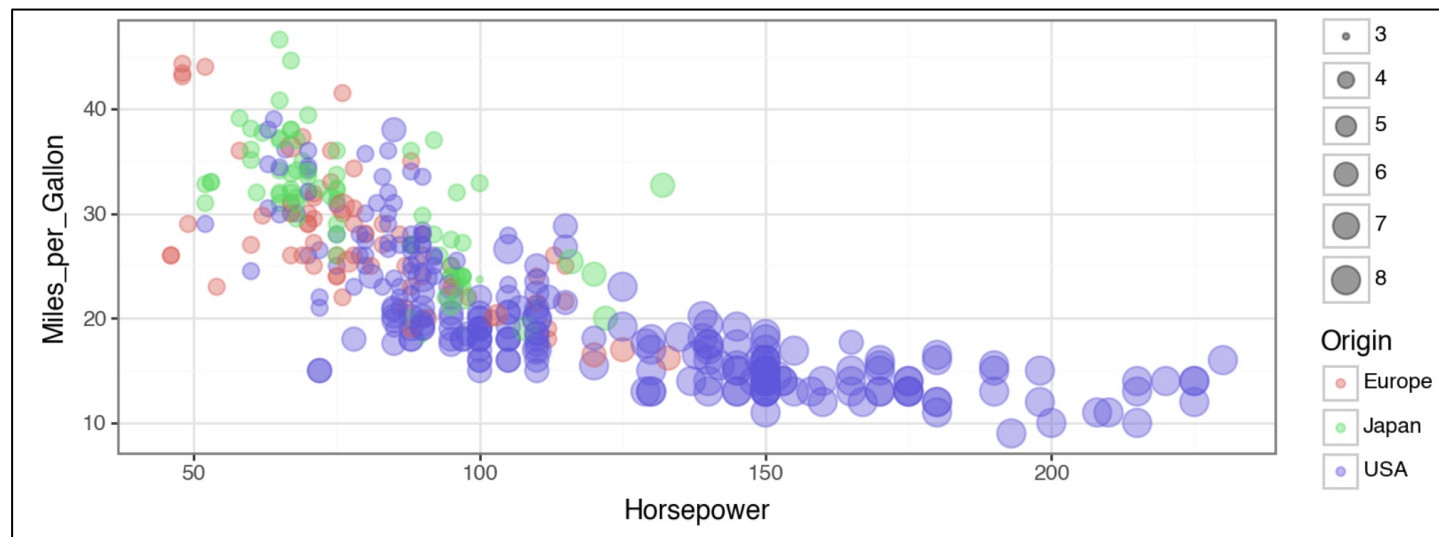


```
alt.Chart(cars).mark_circle(opacity = 0.4).encode(  
  x='Horsepower',  
  y='Miles_per_Gallon',  
  color='Origin',  
  size = 'Cylinders',  
)
```

Map data to visual elements

What are we trying to do in a visualization? Plotnine

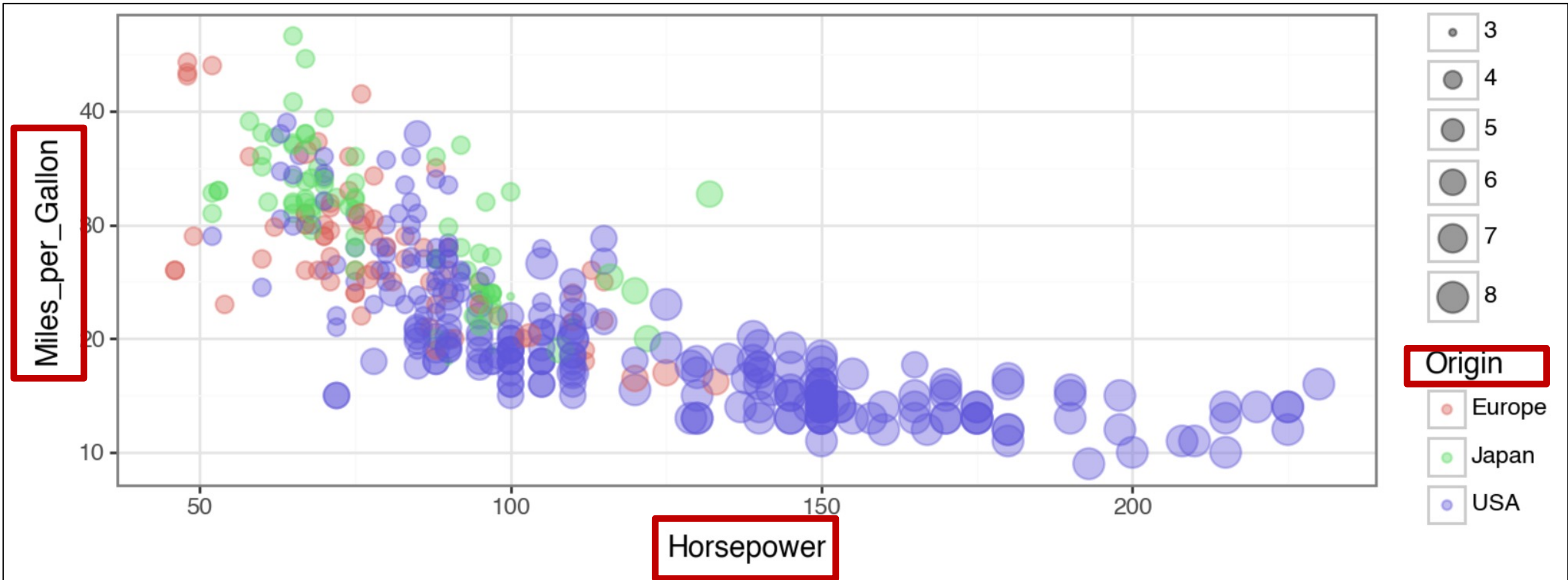
	Name	Miles_per_Gallon	Cylinders	Displacement	Horsepower	Weight_in_lbs	Acceleration	Year	Origin
0	chevrolet chevelle malibu	18.0	8	307.0	130.0	3504	12.0	1970-01-01	USA
1	buick skylark 320	15.0	8	350.0	165.0	3693	11.5	1970-01-01	USA
2	plymouth satellite	18.0	8	318.0	150.0	3436	11.0	1970-01-01	USA
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4	ford torino	17.0	8	302.0	140.0	3449	10.5	1970-01-01	USA
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6	chevrolet impala	14.0	8	454.0	220.0	4354	9.0	1970-01-01	USA
7	plymouth fury iii	14.0	8	440.0	215.0	4312	8.5	1970-01-01	USA
8	pontiac catalina	14.0	8	455.0	225.0	4425	10.0	1970-01-01	USA
9	amc ambassador dpl	15.0	8	390.0	190.0	3850	8.5	1970-01-01	USA
10	citroen ds-21 pallas	NaN	4	133.0	115.0	3090	17.5	1970-01-01	Europe
11	chevrolet chevelle concours (sw)	NaN	8	350.0	165.0	4142	11.5	1970-01-01	USA
12	ford torino (sw)	NaN	8	351.0	153.0	4034	11.0	1970-01-01	USA
13	plymouth satellite (sw)	NaN	8	383.0	175.0	4166	10.5	1970-01-01	USA
14	amc rebel sst (sw)	NaN	8	360.0	175.0	3850	11.0	1970-01-01	USA
15	dodge challenger se	15.0	8	383.0	170.0	3563	10.0	1970-01-01	USA
16	plymouth 'cuda 340	14.0	8	340.0	160.0	3609	8.0	1970-01-01	USA
17	ford mustang boss 302	NaN	8	302.0	140.0	3353	8.0	1970-01-01	USA
18	chevrolet monte carlo	15.0	8	400.0	150.0	3761	9.5	1970-01-01	USA
19	buick estate wagon (sw)	14.0	8	455.0	225.0	3086	10.0	1970-01-01	USA



```
ggplot(cars)
  + aes(    x = 'Horsepower',
            y = 'Miles_per_Gallon',
            color = 'Origin')
  + geom_point(aes(size='Cylinders'), alpha=0.4)
  + theme_bw() + theme(figure_size=(8, 3))
```

Map data to visual elements

Map "data" to "visual elements"



Spend some time with these two papers (week 02a of the schedule)

How to Read a Paper

Version of February 17, 2016

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ABSTRACT

Researchers spend a great deal of time reading research papers. However, this skill is rarely taught, leading to much wasted effort. This article outlines a practical and efficient *three-pass method* for reading research papers. I also describe how to use this method to do a literature survey.

1. INTRODUCTION

Researchers must read papers for several reasons: to review them for a conference or a class, to keep current in their field, or for a literature survey of a new field. A typical researcher will likely spend hundreds of hours every year reading papers.

Learning to efficiently read a paper is a critical but rarely taught skill. Beginning graduate students, therefore, must learn on their own using trial and error. Students waste much effort in the process and are frequently driven to frustration.

For many years I have used a simple ‘three-pass’ approach to prevent me from drowning in the details of a paper before getting a bird’s-eye-view. It allows me to estimate the amount of time required to review a set of papers. Moreover, I can adjust the depth of paper evaluation depending on my needs and how much time I have. This paper describes the approach and its use in doing a literature survey.

2. THE THREE-PASS APPROACH

The key idea is that you should read the paper in up to three passes, instead of starting at the beginning and plowing your way to the end. Each pass accomplishes specific goals and builds upon the previous pass: The *first* pass gives you a general idea about the paper. The *second* pass lets you grasp the paper’s content, but not its details. The *third* pass helps you understand the paper in depth.

2.1 The first pass

4. Read the conclusions

5. Glance over the references, mentally ticking off the ones you’ve already read

At the end of the first pass, you should be able to answer the *five Cs*:

1. *Category*: What type of paper is this? A measurement paper? An analysis of an existing system? A description of a research prototype?

2. *Context*: Which other papers is it related to? Which theoretical bases were used to analyze the problem?

3. *Correctness*: Do the assumptions appear to be valid?

4. *Contributions*: What are the paper’s main contributions?

5. *Clarity*: Is the paper well written?

Using this information, you may choose not to read further (and not print it out, thus saving trees). This could be because the paper doesn’t interest you, or you don’t know enough about the area to understand the paper, or that the authors make invalid assumptions. The first pass is adequate for papers that aren’t in your research area, but may someday prove relevant.

Incidentally, when you write a paper, you can expect most reviewers (and readers) to make only one pass over it. Take care to choose coherent section and sub-section titles and to write concise and comprehensive abstracts. If a reviewer cannot understand the gist after one pass, the paper will likely be rejected; if a reader cannot understand the highlights of the paper after five minutes, the paper will likely never be read. For these reasons, a ‘graphical abstract’ that summarizes a paper with a single well-chosen figure is an excellent idea and can be increasingly found in scientific journals.



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MMMMMM YYYY, Volume VV, Issue II.

<http://www.jstatsoft.org/>

Tidy Data

Hadley Wickham
RStudio

Abstract

A huge amount of effort is spent cleaning data to get it ready for analysis, but there has been little research on how to make data cleaning as easy and effective as possible. This paper tackles a small, but important, component of data cleaning: data tidying. Tidy datasets are easy to manipulate, model and visualise, and have a specific structure: each variable is a column, each observation is a row, and each type of observational unit is a table. This framework makes it easy to tidy messy datasets because only a small set of tools are needed to deal with a wide range of un-tidy datasets. This structure also makes it easier to develop tidy tools for data analysis, tools that both input and output tidy datasets. The advantages of a consistent data structure and matching tools are demonstrated with a case study free from mundane data manipulation chores.

Keywords: data cleaning, data tidying, relational databases, R.

Extracting Variables and Observations are easier in Tidy Data

Country	2011	2012	2013
FR	7000	6900	7000
DE	5800	6000	6200
US	15000	14000	13000

messy

```
country = df.country  
year = df.columns[1:]  
count = df.values
```

```
country = df.country  
year = df.year  
count = df.count
```

Country	Year	Count
FR	2011	7000
DE	2011	5800
US	2011	15000
FR	2012	6900
DE	2012	6000
US	2012	14000
FR	2013	7000
DE	2013	6200
US	2013	13000

tidy

PANDAS FUNCTIONS FOR TIDYING

melt, pivot_table, stack, unstack

pd.melt

```
pd.melt(  
    dataframe,  
    countries,  
    id_vars=['country'],  
    value_vars=['2011', '2012', '2013'],  
    var_name='year',  
    value_name='count')
```

dataframe

Existing variables
already in columns

Values of the new
variable

Name of the new
variable

Name of the new
value column

countries				
	country	2011	2012	2013
0	FR	7000	6900	7000
1	DE	5800	6000	6200
2	US	15000	14000	13000

	country	year	count
0	FR	2011	7000
1	DE	2011	5800
2	US	2011	15000
3	FR	2012	6900
4	DE	2012	6000
5	US	2012	14000
6	FR	2013	7000
7	DE	2013	6200
8	US	2013	13000

Column headers are values; not variable names

pd.pivot / pd.pivot_table

	e1	e2	e3
Bob			
Jack			
Jill			

	e1	e2	e3
Bob	97	-	99
Jack	92, 98	83	-
Jill	96	-	86 97

	Name	Exam	Score
0	Jack	e1	92
1	Jill	e1	96
2	Jack	e2	83
3	Bob	e1	97
4	Jill	e3	86
5	Jack	e1	98
6	Bob	e3	99
7	Jill	e3	97

columns

```
pd.pivot_table(  
    df,  
    index='name',  
    columns='exam',  
    values='score',  
    aggfunc = lambda s: s.mean()  
)
```

)

0	92
5	98

index

	e1	e2	e3
Bob	97	NaN	99
Jack	95	83	NaN
Jill	96	NaN	91.5

pd.stack / unstack

```
df.unstack()  
df.unstack(-1)  
df.unstack(1)
```

		SAT	ACT
PA	Jack	1440	33
PA	Jill	1550	36
FL	Bob	1340	30
IN	Harry	1440	34
IN	Pat	1530	35
IN	Carol	1490	31

	sat						act					
name	Bob	Carol	Harry	Jack	Jill	Pat	Bob	Carol	Harry	Jack	Jill	Pat
state												
FL	1340	NaN	NaN	NaN	NaN	NaN	30	NaN	NaN	NaN	NaN	NaN
IN	NaN	1490	1440	NaN	NaN	1530	NaN	31	34	NaN	NaN	35
PA	NaN	NaN	NaN	1440	1550	NaN	NaN	NaN	NaN	33	36	NaN

```
df.unstack(0)
```

	sat			act		
state	FL	IN	PA	FL	IN	PA
name						
Bob	1340	NaN	NaN	30	NaN	NaN
Carol	NaN	1490	NaN	NaN	31	NaN
Harry	NaN	1440	NaN	NaN	34	NaN
Jack	NaN	NaN	1440	NaN	NaN	33
Jill	NaN	NaN	1550	NaN	NaN	36
Pat	NaN	1530	NaN	NaN	35	NaN

Indices are numbered 0, 1, ..., -1

Five Reasons for Untidyness

1. Column headers are values, not variable names
2. Multiple variables are stored in one column
3. Variables are stored in both rows and columns
4. Multiple types of observational units are stored in the same table
5. A single observational unit is stored in multiple tables

Following examples are from Hadley Wickham's paper on [Tidy Data](#)

Happy families are all alike;
every unhappy family is unhappy in its own way
— Leo Tolstoy

Full version of the data

	religion	<\$10k	\$10-20k	\$20-30k	\$30-40k	\$40-50k	\$50-75k	\$75-100k	\$100-150k	>150k	Don't know/refused
0	Agnostic	27	34	60	81	76	137	122	109	84	96
1	Atheist	12	27	37	52	35	70	73	59	74	76
2	Buddhist	27	21	30	34	33	58	62	39	53	54
3	Catholic	418	617	732	670	638	1116	949	792	633	1489
4	Don't know/refused	15	14	15	11	10	35	21	17	18	116
5	Evangelical Prot	575	869	1064	982	881	1486	949	723	414	1529
6	Hindu	1	9	7	9	11	34	47	48	54	37
7	Historically Black Prot	228	244	236	238	197	223	131	81	78	339
8	Jehovah's Witness	20	27	24	24	21	30	15	11	6	37
9	Jewish	19	19	25	25	30	95	69	87	151	162
10	Mainline Prot	289	495	619	655	651	1107	939	753	634	1328
11	Mormon	29	40	48	51	56	112	85	49	42	69
12	Muslim	6	7	9	10	9	23	16	8	6	22
13	Orthodox	13	17	23	32	32	47	38	42	46	73
14	Other Christian	9	7	11	13	13	14	18	14	12	18
15	Other Faiths	20	33	40	46	49	63	46	40	41	71
16	Other World Religions	5	2	3	4	2	7	3	4	4	8
17	Unaffiliated	217	299	374	365	341	528	407	321	258	597

1. Column headers are values, not variable names 1/4

	<\$10k	\$10-20k	\$20-30k	\$30-40k	\$40-50k	\$50-75k
religion						
Agnostic	27	34	60	81	76	137
Atheist	12	27	37	52	35	70
Buddhist	27	21	30	34	33	58
Catholic	418	617	732	670	638	1116
Dont know/refused	15	14	15	11	10	35
Evangelical Prot	575	869	1064	982	881	1486
Hindu	1	9	7	9	11	34
Historically Black Prot	228	244	236	238	197	223
Jehovahs Witness	20	27	24	24	21	30
Jewish	19	19	25	25	30	95

What are the variables?

What are the observations?

1. Column headers are values, not variable names 2/4

		<\$10k	\$10-20k	\$20-30k	\$30-40k	\$40-50k	\$50-75k
religion							
Agnostic		27	34	60	81	76	137
Atheist		12	27	37	52	35	70
Buddhist		27	21	30	34	33	58
Catholic		418	617	732	670	638	1116
Dont know/refused		15	14	15	11	10	35
Evangelical Prot		575	869	1064	982	881	1486
Hindu		1	9	7	9	11	34
Historically Black Prot		228	244	236	238	197	223
Jehovahs Witness		20	27	24	24	21	30
Jewish		19	19	25	25	30	95

What are the variables?

1. Column headers are values, not variable names 3/4

	<\$10k	\$10-20k	\$20-30k	\$30-40k	\$40-50k	\$50-75k
religion						
Agnostic	27	34	60	81	76	137
Atheist	12	27	37	52	35	70
Buddhist	27	21	30	34	33	58
Catholic	418	617	732	670	638	1116
Dont know/refused	15	14	15	11	10	35
Evangelical Prot	575	869	1064	982	881	1486
Hindu	1	9	7	9	11	34
Historically Black Prot	228	244	236	238	197	223
Jehovahs Witness	20	27	24	24	21	30
Jewish	19	19	25	25	30	95

What are the observations?

1. Column headers are values, not variable names 4/4

10 x 6

	<\$10k	\$10-20k	\$20-30k	\$30-40k	\$40-50k	\$50-75k
religion						
Agnostic	27	34	60	81	76	137
Atheist	12	27	37	52	35	70
Buddhist	27	21	30	34	33	58
Catholic	418	617	732	670	638	1116
Dont know/refused	15	14	15	11	10	35
Evangelical Prot	575	869	1064	982	881	1486
Hindu	1	9	7	9	11	34
Historically Black Prot	228	244	236	238	197	223
Jehovahs Witness	20	27	24	24	21	30
Jewish	19	19	25	25	30	95

60 x 3

religion	income	freq
Agnostic	<\$10k	27
Agnostic	\$30-40k	81
Agnostic	\$40-50k	76
Agnostic	\$50-75k	137
Agnostic	\$10-20k	34
Agnostic	\$20-30k	60
Atheist	\$40-50k	35
Atheist	\$20-30k	37
Atheist	\$10-20k	27
Atheist	\$30-40k	52

EFFECTIVE VISUALIZATIONS

- *Examples of effective visualizations*
- *Theory of Visualization*
- *Introduction to Altair*

OVER 15 MILLION SOLD

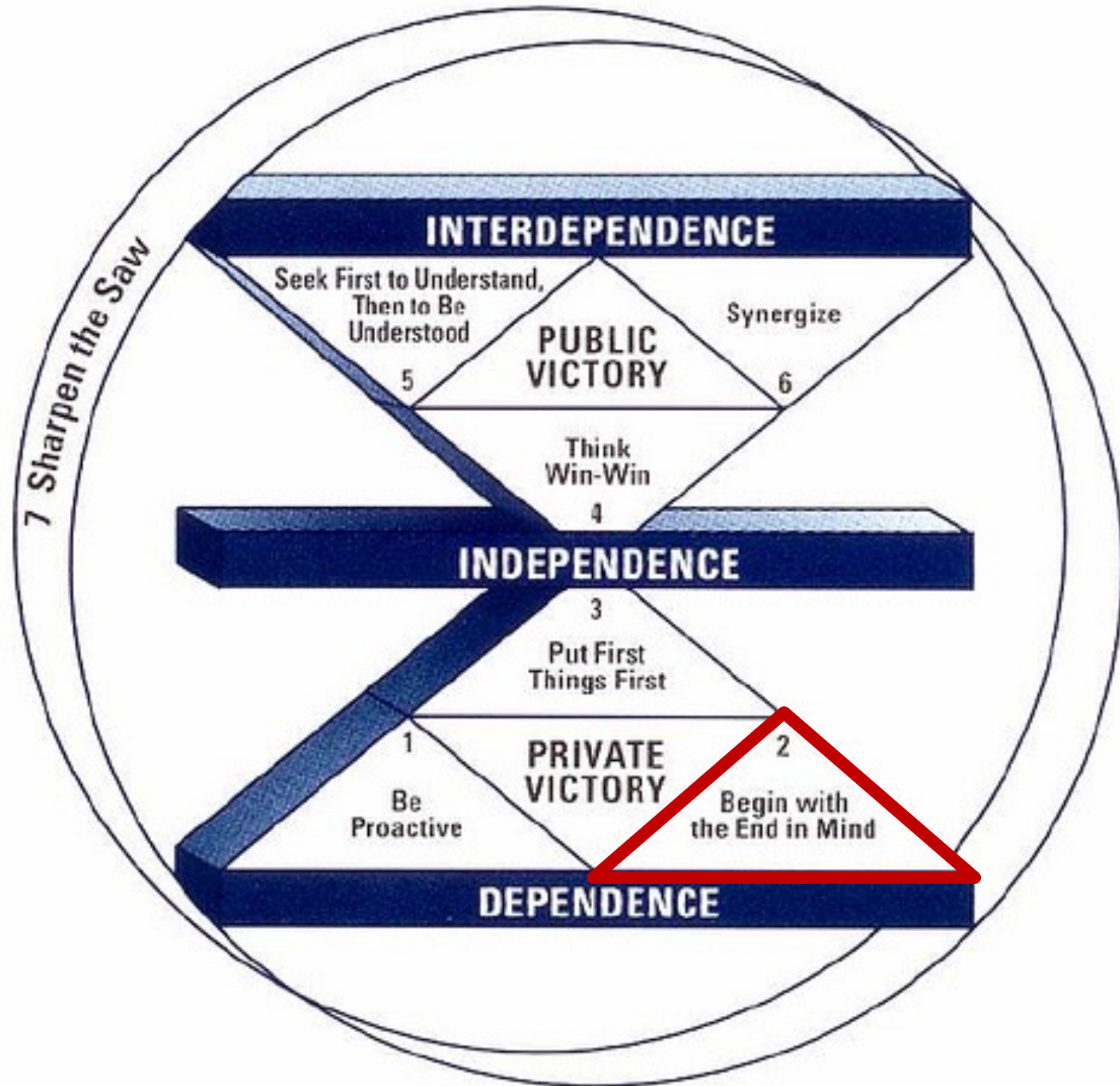
THE 7 HABITS OF HIGHLY EFFECTIVE PEOPLE

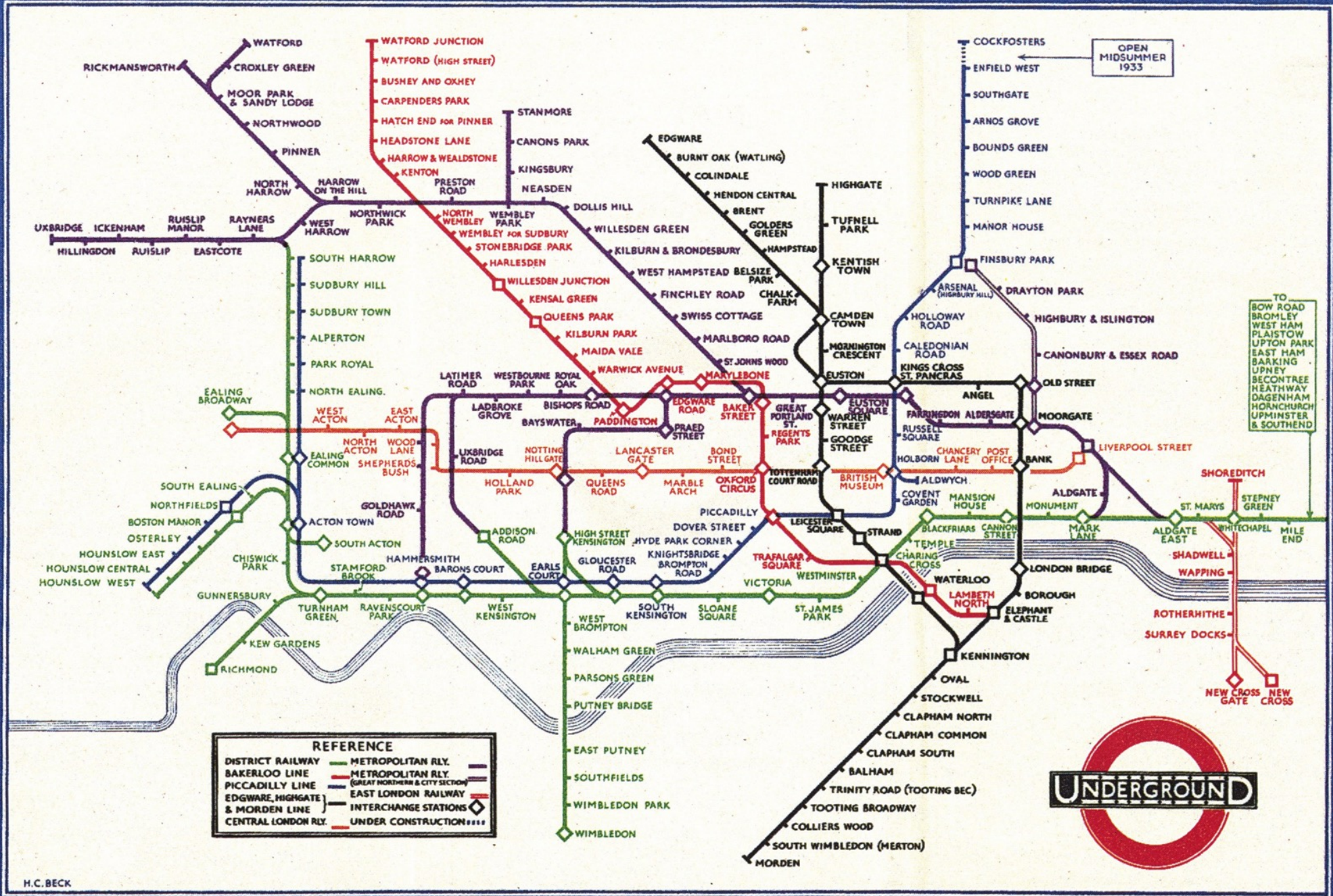
Powerful Lessons
in Personal Change

With a New
Foreword and
Afterword
by the Author

"A wonderful book that could change your life."
—Tom Peters, bestselling author of *In Search of Excellence*

Stephen R. Covey



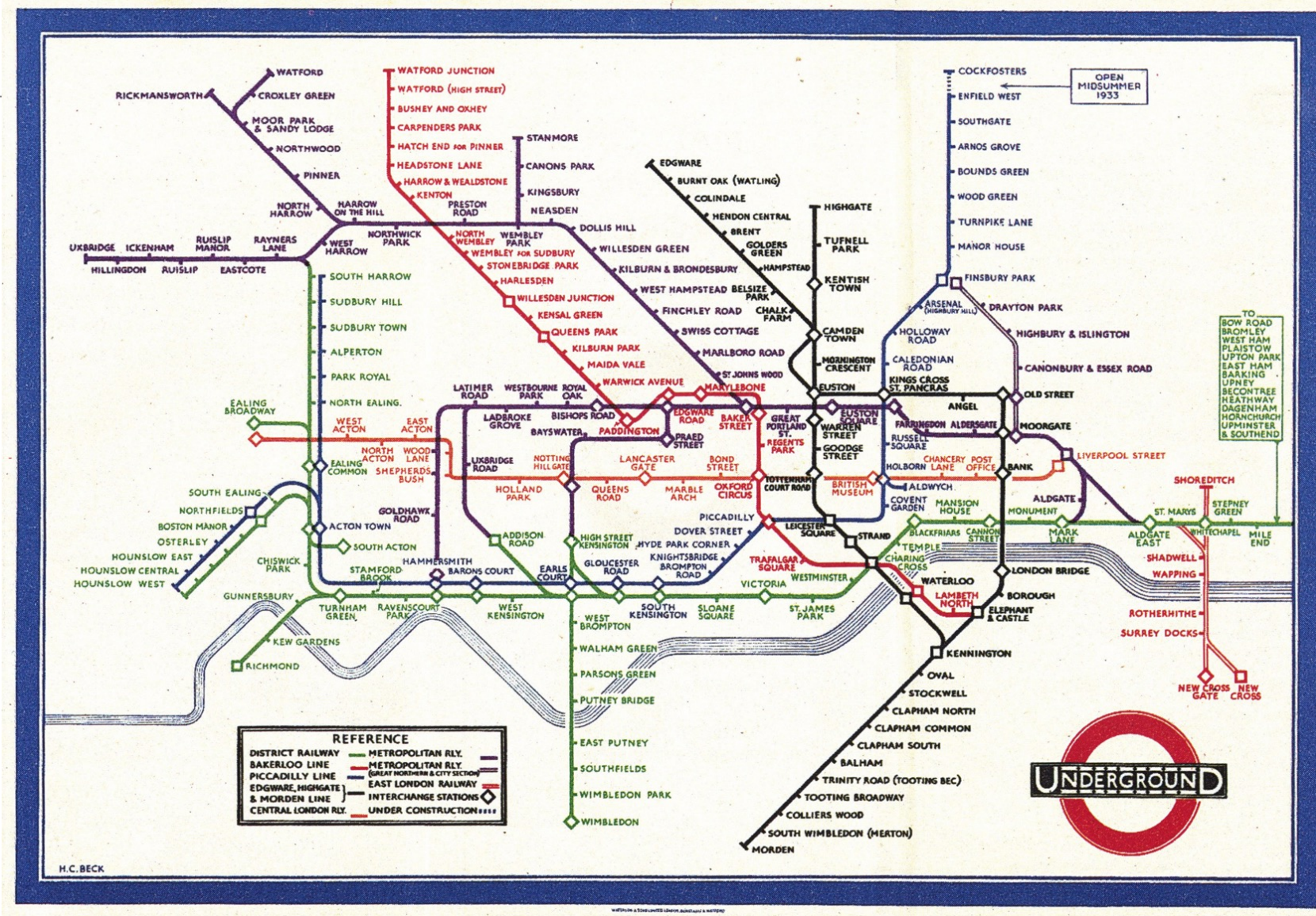


REFERENCE	
DISTRICT RAILWAY	METROPOLITAN RLY.
BAKERLOO LINE	METROPOLITAN RLY.
PICCADILLY LINE	GREAT NORTHERN & CITY SECTION
EDGWARE, HIGHGATE & MORDEN LINE	EAST LONDON RAILWAY
	INTERCHANGE STATIONS
CENTRAL LONDON RLY.	UNDER CONSTRUCTION

UNDERGROUND

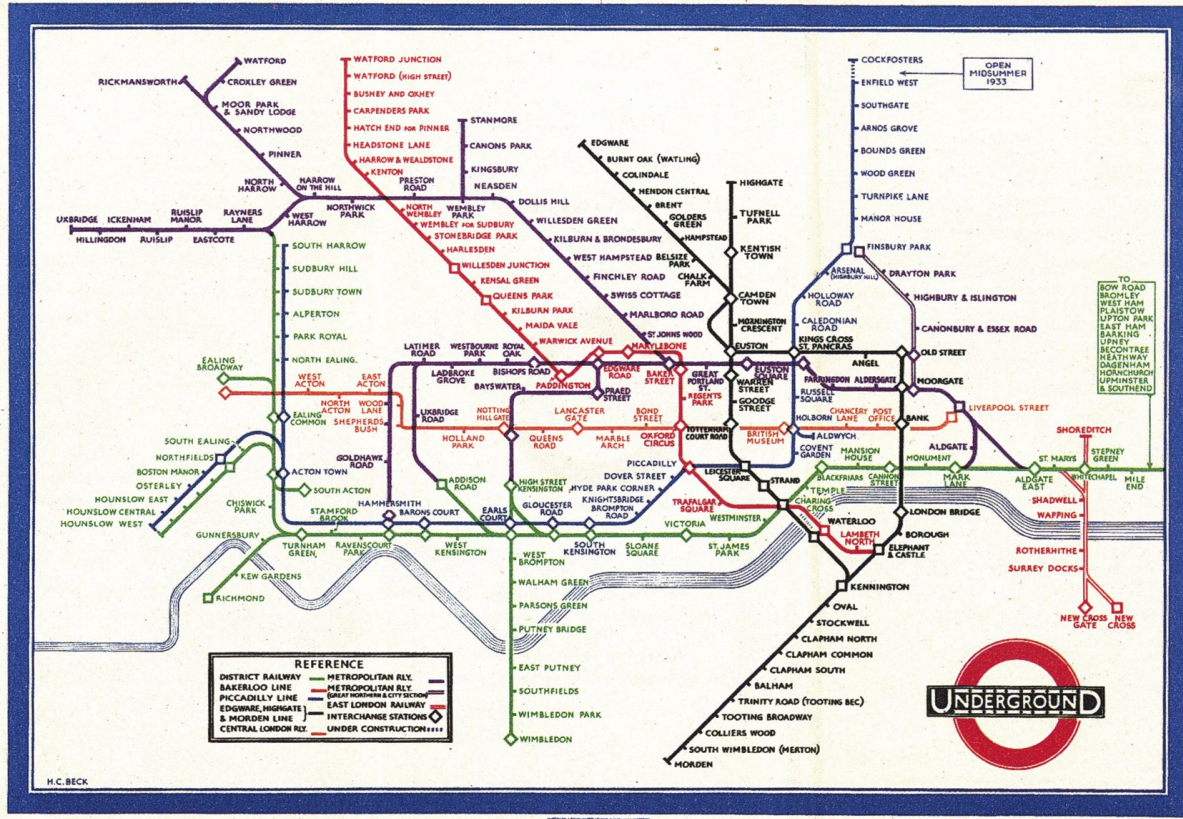
RAILWAYS





User's Task: Understand how to get from point A to point B

Important Information: Sequence of stops / interchanges along each line

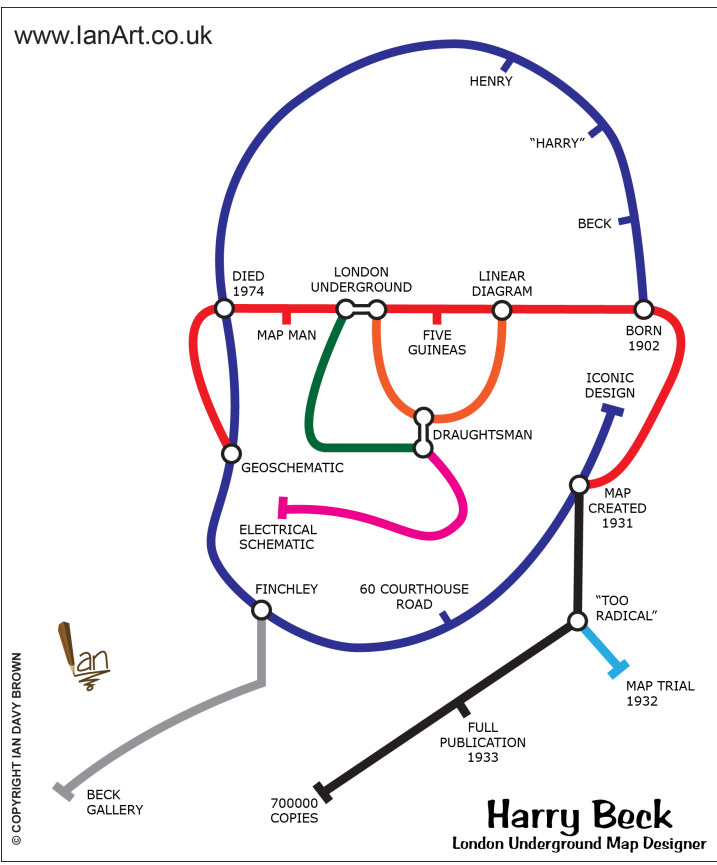
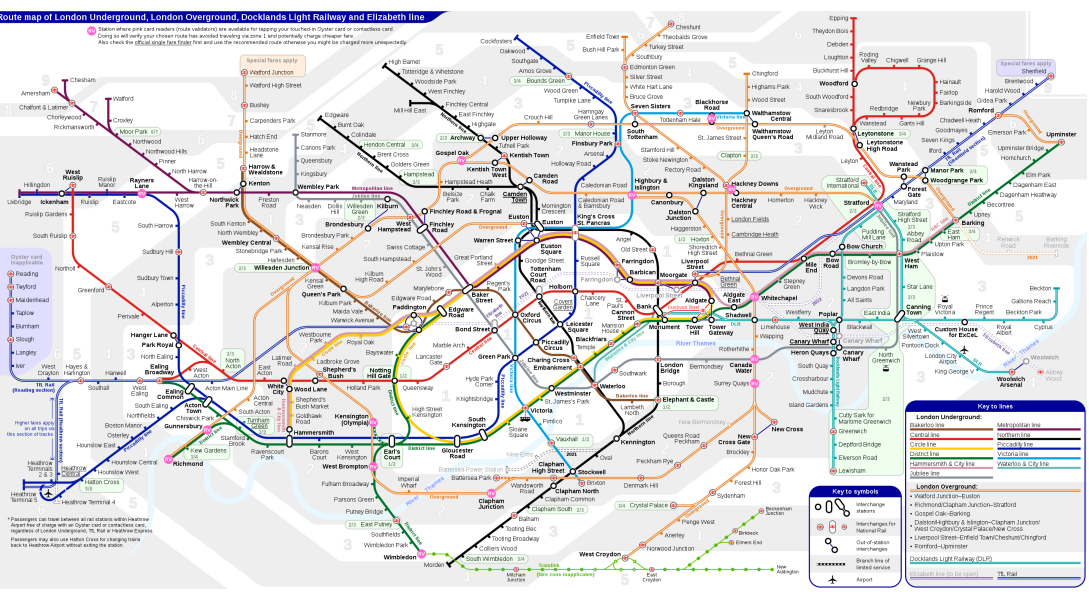


Design Principles:

Straight lines and evenly spaced stops to emphasize sequence of steps
De-emphasize geographic shape of subway lines

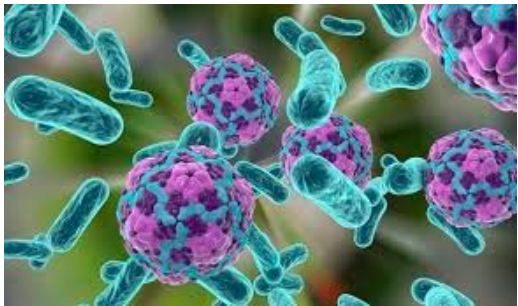
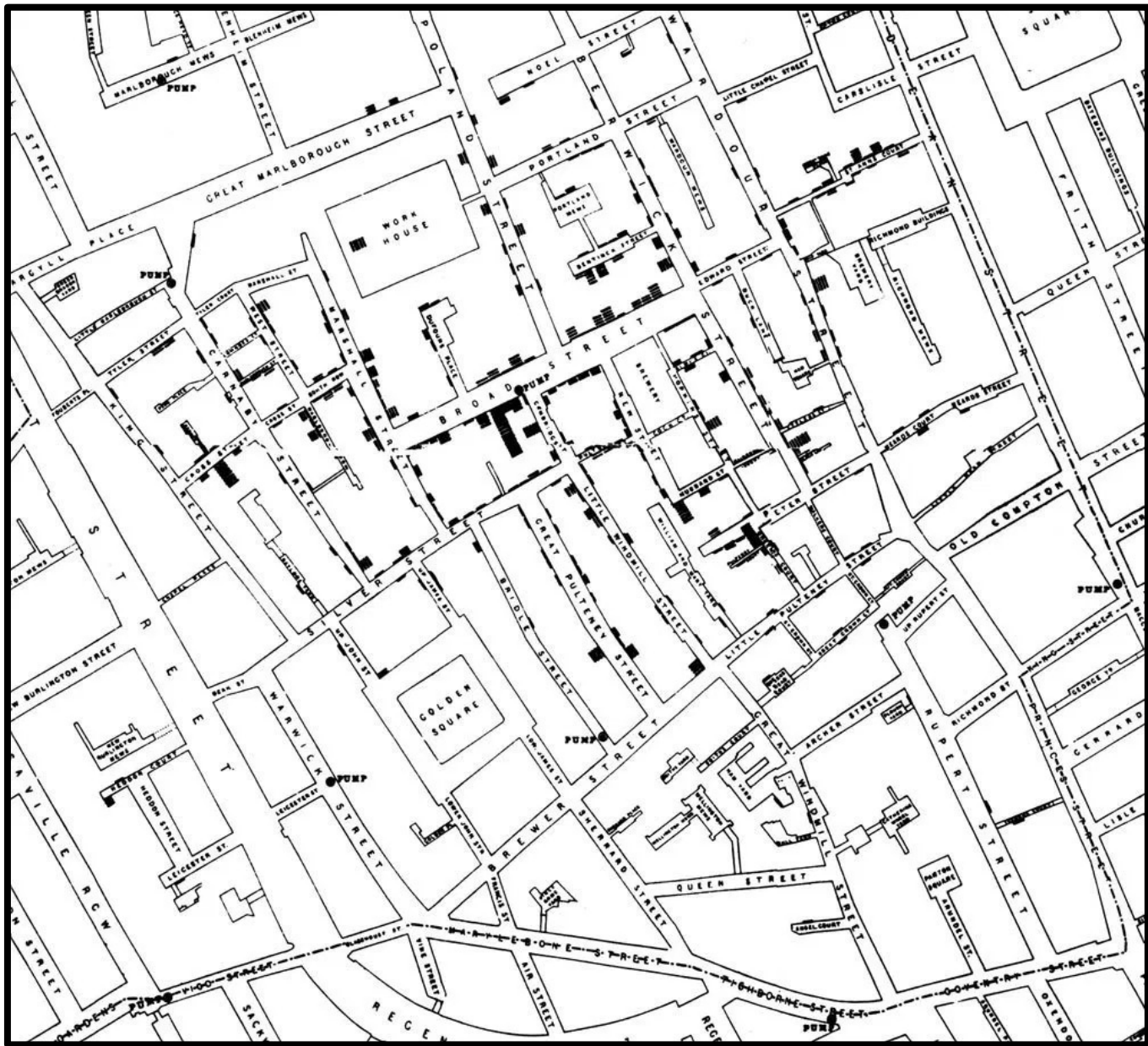
Harry Beck

Henry Charles Beck (4 June 1902 – 18 September 1974) was an English technical draughtsman who created the present London Underground Tube map in 1931. Beck drew the diagram in his spare time while working at the London Underground Signals Office. London Underground tentatively introduced it to the public in a small pamphlet in 1933. It was immediately popular, and the Underground has used topological maps to illustrate the network ever since.

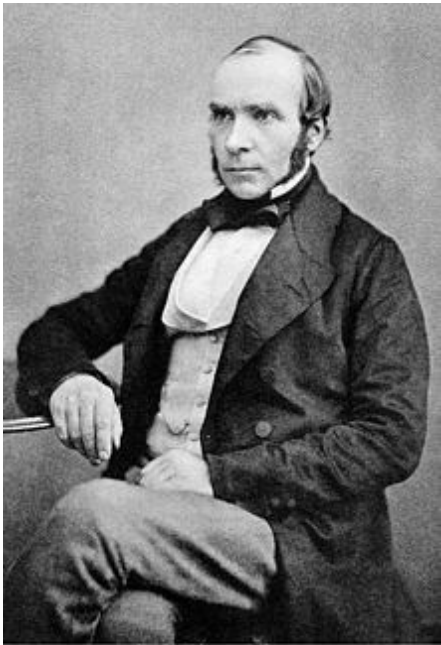


THE GHOST MAP

Soho, London 1854

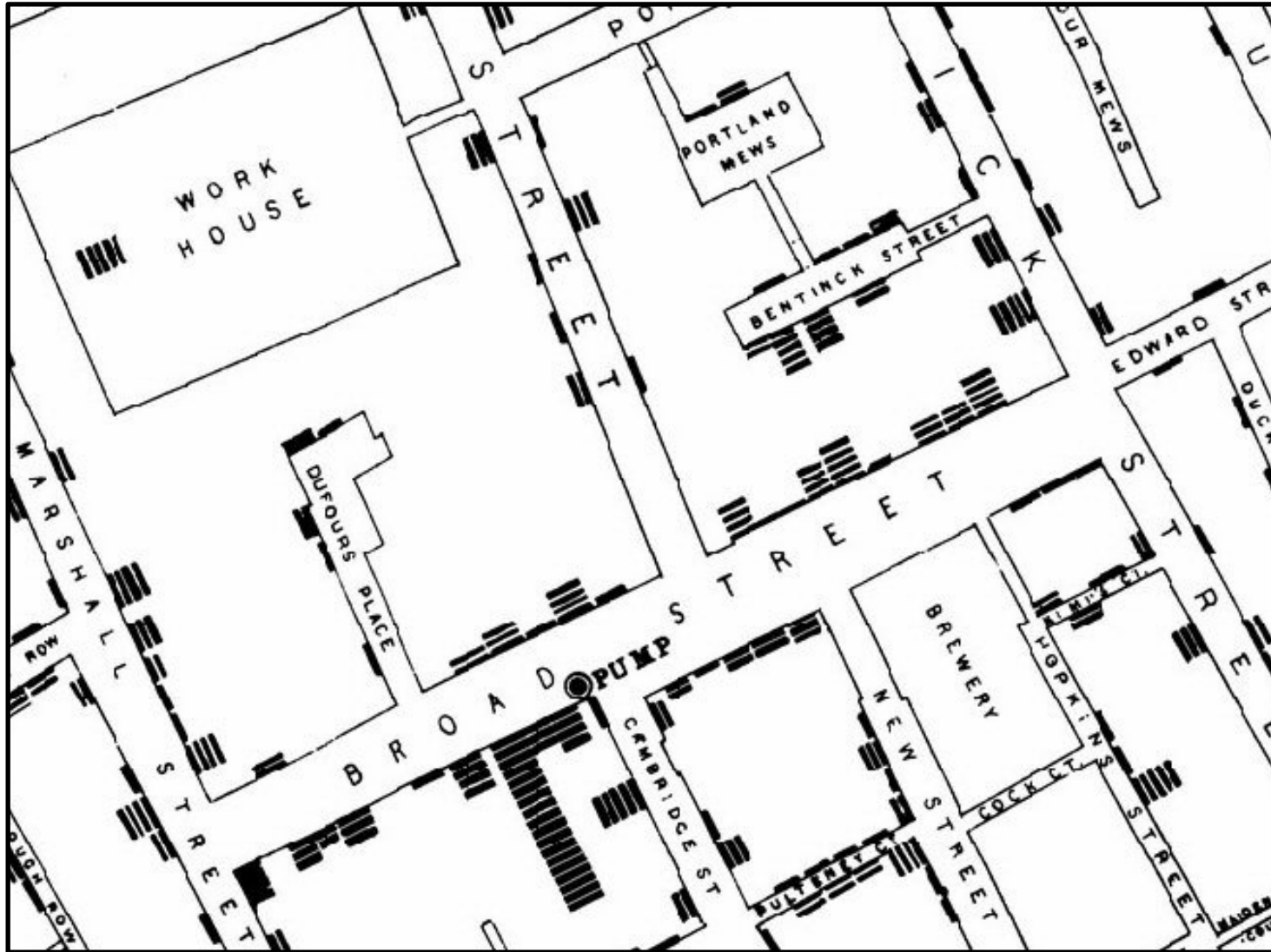


Cholera germs



John Snow

The Broad Street Pump



MOVIE LENS CASE STUDY

File	Fields
u.data (10000)	user id movie id rating timestamp
u.item (1682)	movie id movie title release date video release date IMDB URL unknown Action Adventure Animation Children's Comedy Crime Documentary Drama Fantasy Film-Noir Horror Musical Mystery Romance Sci-Fi Thriller War Western
u.User (102625)	user id age gender occupation zip code

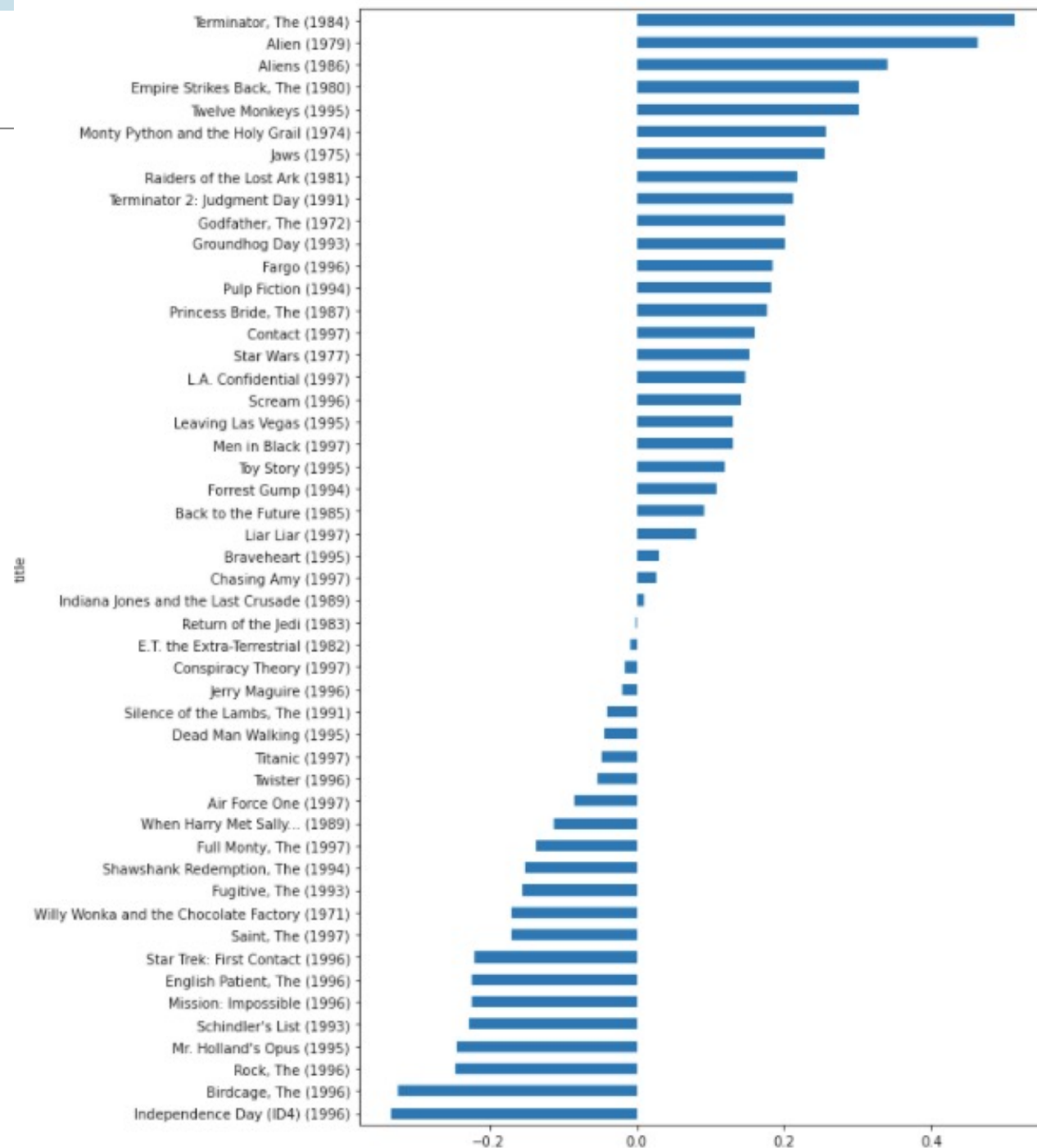
Question: Given the above files (fields) how would you conduct the analysis to answer the question:
How do men and women rate movies differently? Try to answer the following questions on paper:

1. How would you combine the data in the 3 tables into a single table? What fields do you need?
2. How would you analyze the combined table?
3. What would be the data format (tabular format) of the final analysis?
4. How would you visually convey your final analysis?

1. What would be the tabular format of the final analysis?

Movie_id	Movie_title	Male_rating	Female_rating

Movie_id	Movie_title	Male_rating	Female_rating	Diff





Stanley Smith Stevens



Jacques Bertin

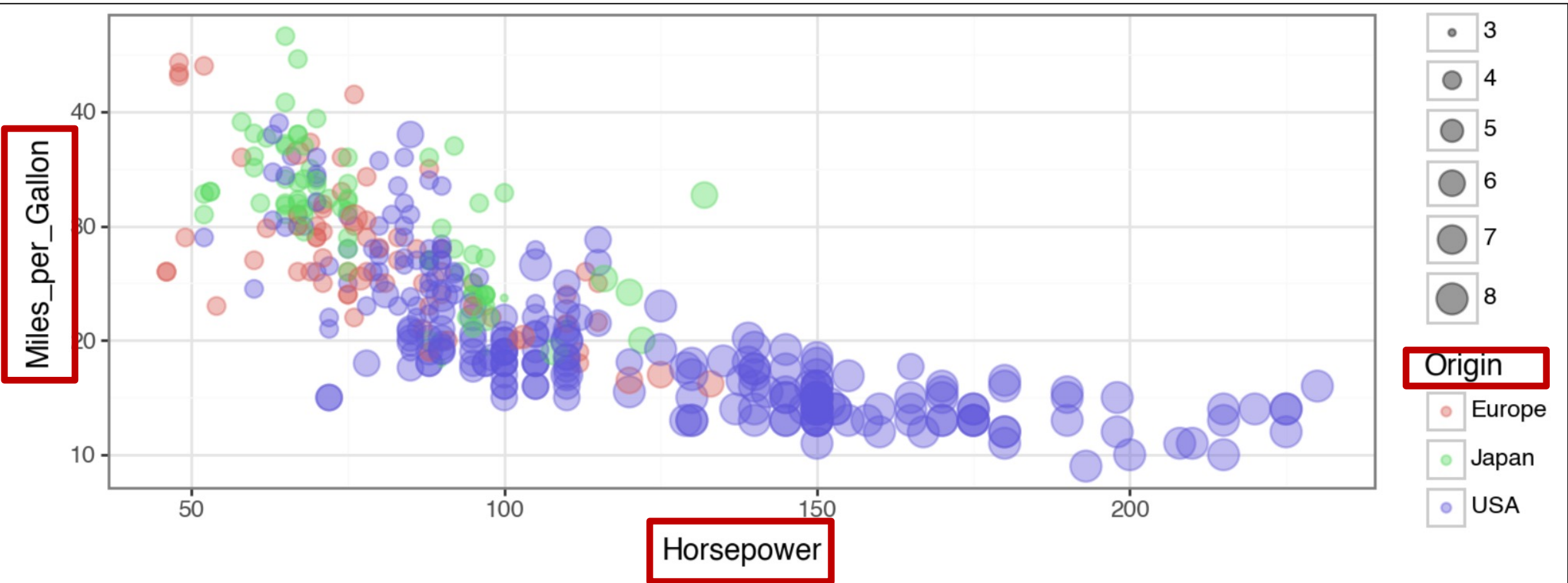


Leland Wilkinson

EFFECTIVE MAPPING

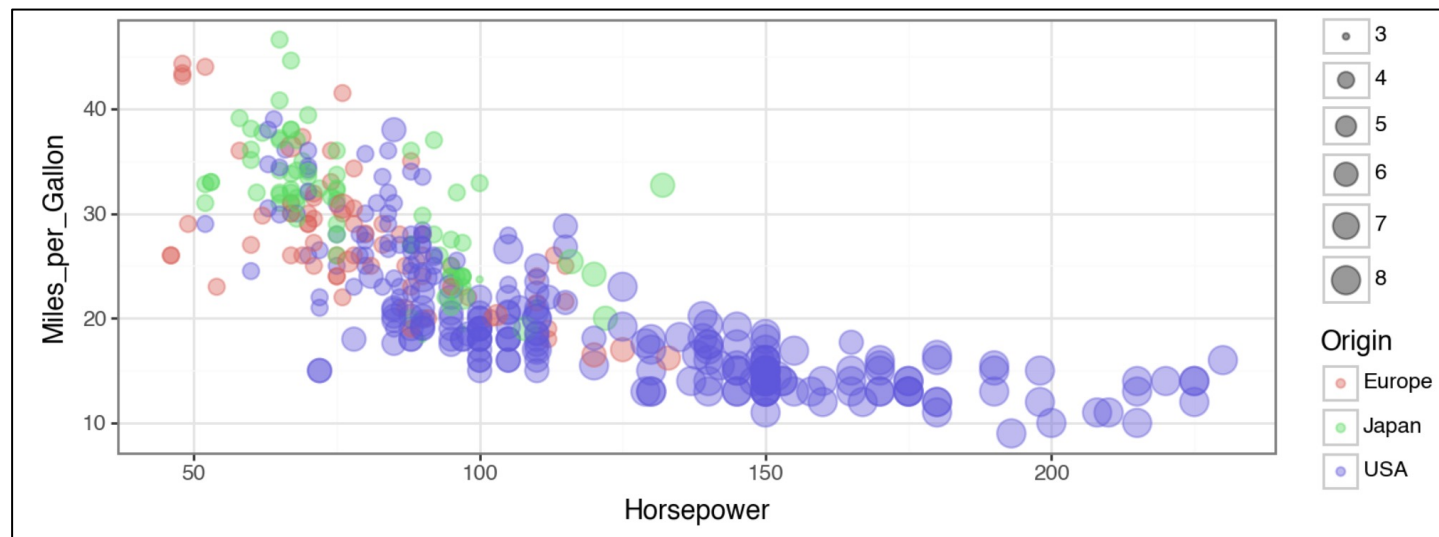
Formalizing the practice: Stevens, Bertin, Wilkinson

Map "data" to "visual elements"



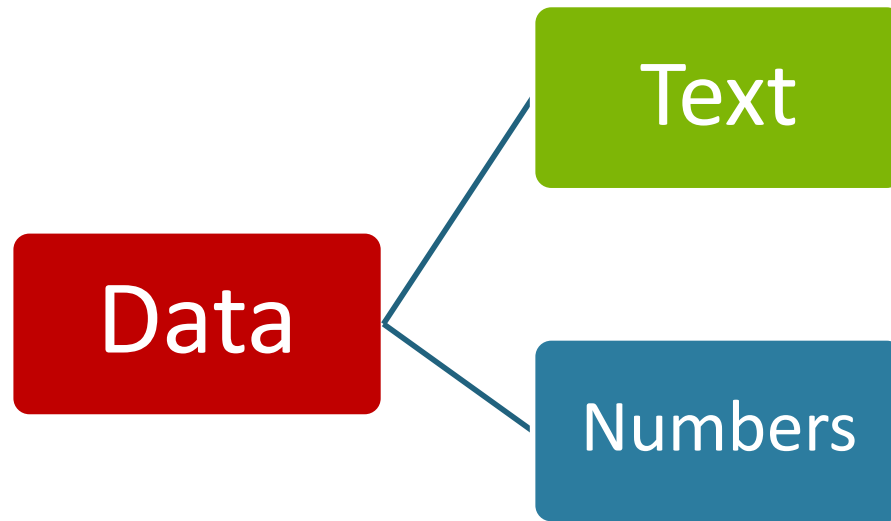
What are we trying to do in a visualization? Plotnine

	Name	Miles_per_Gallon	Cylinders	Displacement	Horsepower	Weight_in_lbs	Acceleration	Year	Origin
0	chevrolet chevelle malibu	18.0	8	307.0	130.0	3504	12.0	1970-01-01	USA
1	buick skylark 320	15.0	8	350.0	165.0	3693	11.5	1970-01-01	USA
2	plymouth satellite	18.0	8	318.0	150.0	3436	11.0	1970-01-01	USA
3	amc rebel sst	16.0	8	304.0	150.0	3433	12.0	1970-01-01	USA
4	ford torino	17.0	8	302.0	140.0	3449	10.5	1970-01-01	USA
5	ford galaxie 500	15.0	8	429.0	198.0	4341	10.0	1970-01-01	USA
6	chevrolet impala	14.0	8	454.0	220.0	4354	9.0	1970-01-01	USA
7	plymouth fury iii	14.0	8	440.0	215.0	4312	8.5	1970-01-01	USA
8	pontiac catalina	14.0	8	455.0	225.0	4425	10.0	1970-01-01	USA
9	amc ambassador dpl	15.0	8	390.0	190.0	3850	8.5	1970-01-01	USA
10	citroen ds-21 pallas	NaN	4	133.0	115.0	3090	17.5	1970-01-01	Europe
11	chevrolet chevelle concours (sw)	NaN	8	350.0	165.0	4142	11.5	1970-01-01	USA
12	ford torino (sw)	NaN	8	351.0	153.0	4034	11.0	1970-01-01	USA
13	plymouth satellite (sw)	NaN	8	383.0	175.0	4166	10.5	1970-01-01	USA
14	amc rebel sst (sw)	NaN	8	360.0	175.0	3850	11.0	1970-01-01	USA
15	dodge challenger se	15.0	8	383.0	170.0	3563	10.0	1970-01-01	USA
16	plymouth 'cuda 340	14.0	8	340.0	160.0	3609	8.0	1970-01-01	USA
17	ford mustang boss 302	NaN	8	302.0	140.0	3353	8.0	1970-01-01	USA
18	chevrolet monte carlo	15.0	8	400.0	150.0	3761	9.5	1970-01-01	USA
19	buick estate wagon (sw)	14.0	8	455.0	225.0	3086	10.0	1970-01-01	USA



```
ggplot(cars)
+ aes(
  x = 'Horsepower',
  y = 'Miles_per_Gallon',
  color = 'Origin')
+ geom_point(aes(size='Cylinders'), alpha=0.4)
+ theme_bw() + theme(figure_size=(8, 3))
```

Map data to visual elements

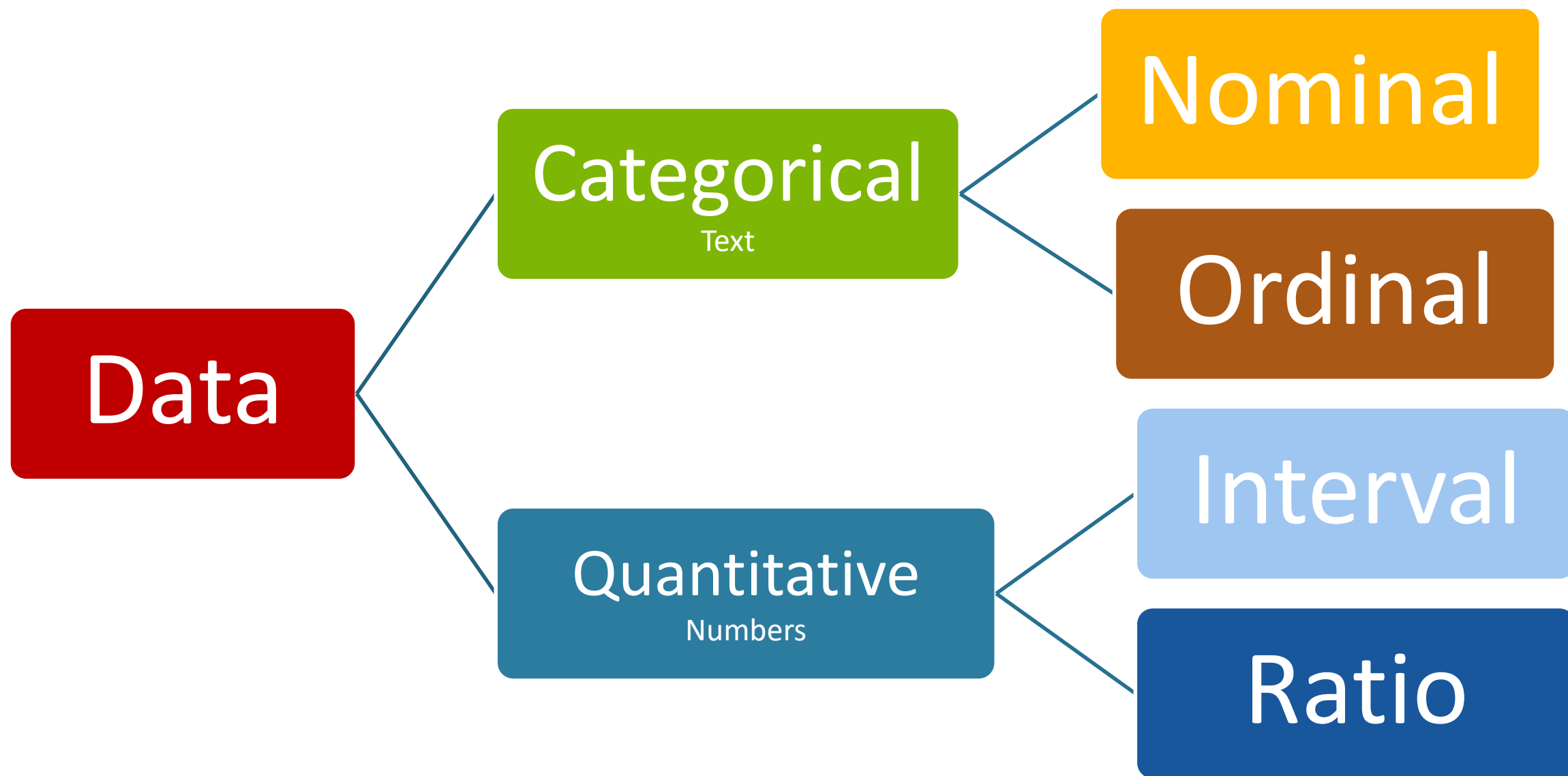


DATA

text

numbers

Types of Data



Types of Data (Stevens)

- **N: Nominal (labels)**

Examples: genotype, blood type, zip code, gender, race, eye color, political party

Operations: = !=

- **O: Ordered**

Examples: socio economic status (“low income”, “middle income”, “high income”), income level (“less than 50K”, “50K-100K”, “over 100K”), satisfaction rating (“extremely dislike”, “dislike”, “neutral”, “like”, “extremely like”).

Operations: = != < > <= >=

- **Q: Interval (location of zero arbitrary)**

Examples: temperature (F or C), pH, SAT score (200-800), credit score (300-850).

Cannot compare directly; only differences (i.e., intervals may be compared)

Operations: = != < > <= >= -

- **Q: Ratio (location of zero fixed)**

Physical measurement (length, mass, temp in K)

Operations: = != < > <= >= - /

NOMINAL (CATEGORICAL)

- Unordered categories
 - Flower species
 - Country names
 - Binary responses



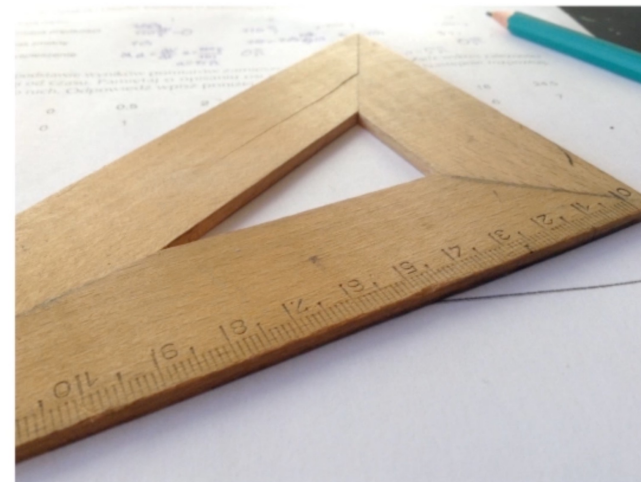
ORDINAL

- Ordered categories
 - Low, medium, high
 - Letter grades (A, B, C)
 - Movie ratings (G, PG, PG-13, etc.)



QUANTITATIVE

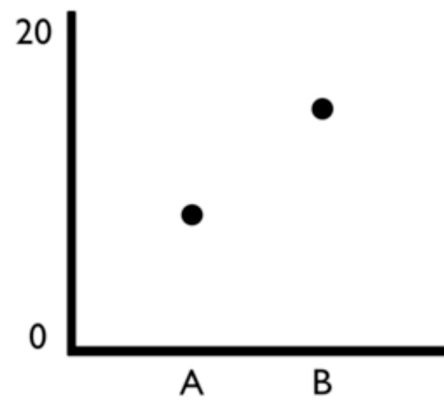
- Measured on a numeric scale
 - Chemical concentrations
 - Temperature
 - Revenue in a given period



Interval
Ratio

VISUAL ENCODINGS

POSITION



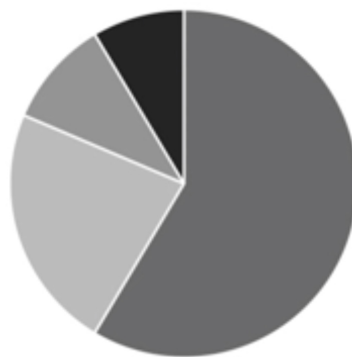
LENGTH



AREA

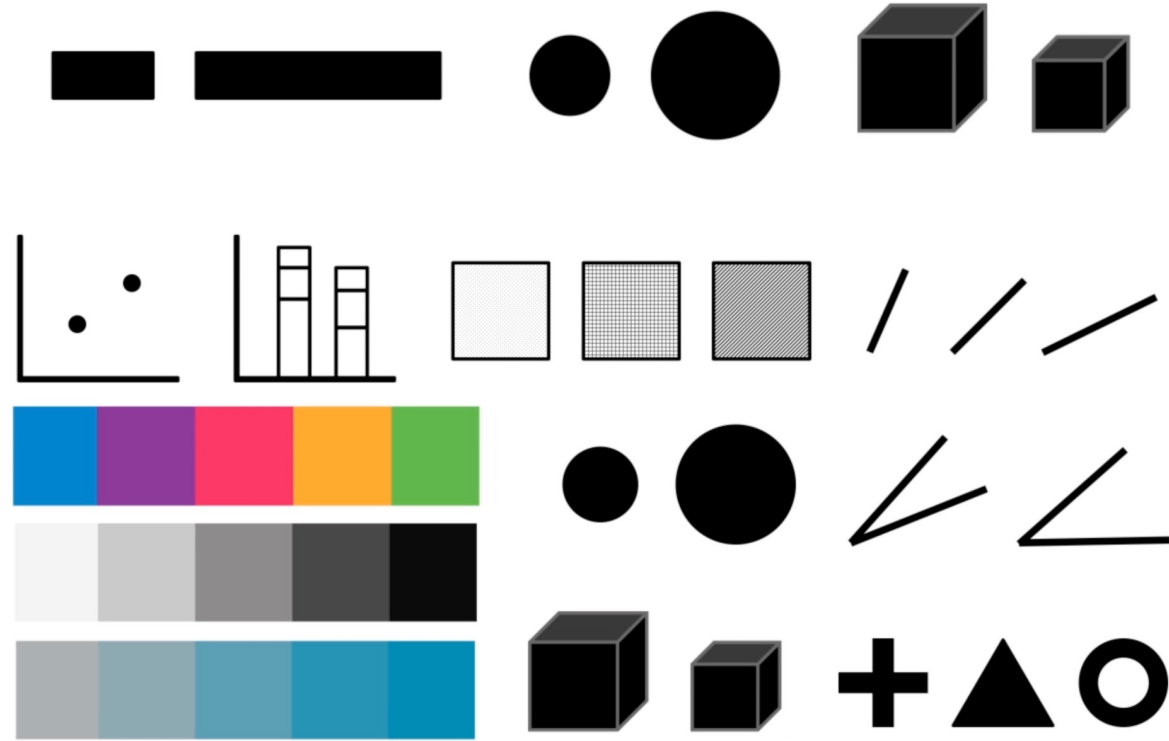


ANGLE



COLOR: SATURATION, LIGHTNESS, HUE

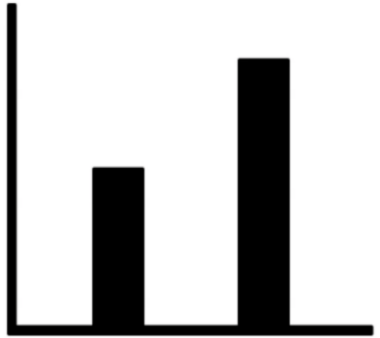




MAPPING DATA TO VISUAL ELEMENTS

Encoding should match data type

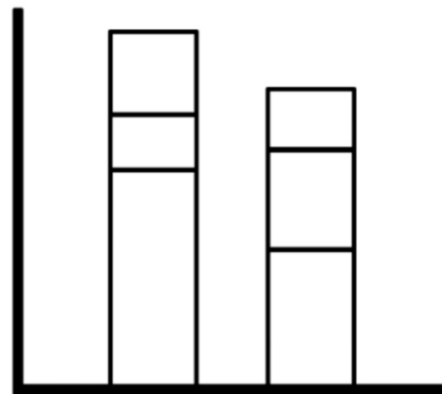
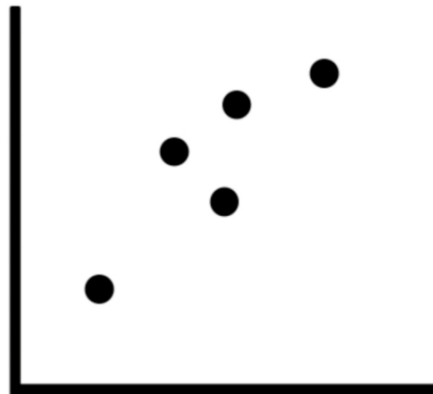
Nominal: discrete, unordered



Ordinal: discrete, ordered



Quantitative: continuous, ordered

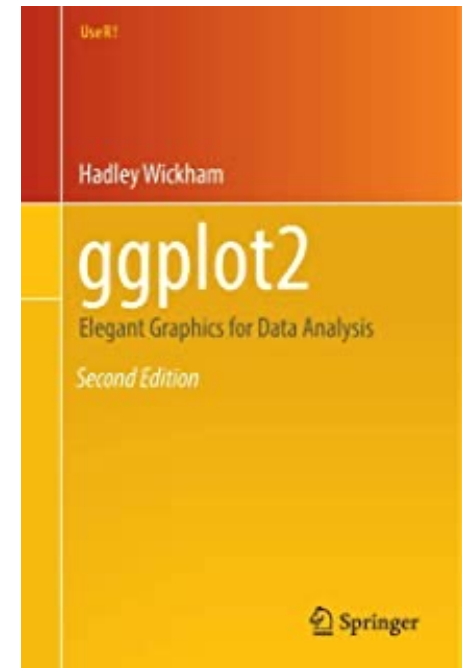
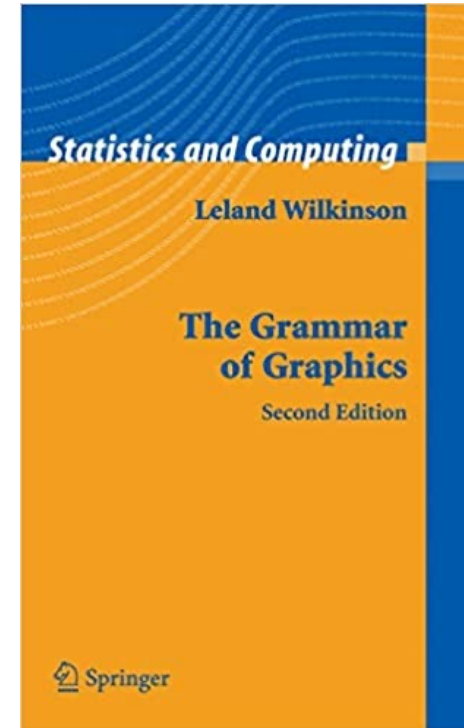


- Nominal: position, hue, texture
- Ordinal: position, lightness, saturation
- Quantitative: position, length, angle, slope

GRAMMAR OF GRAPHICS

The Theory

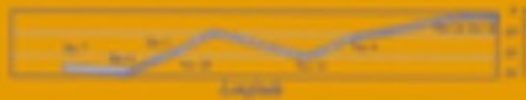
Implementations: ggplot, plotnine, altair, plotly



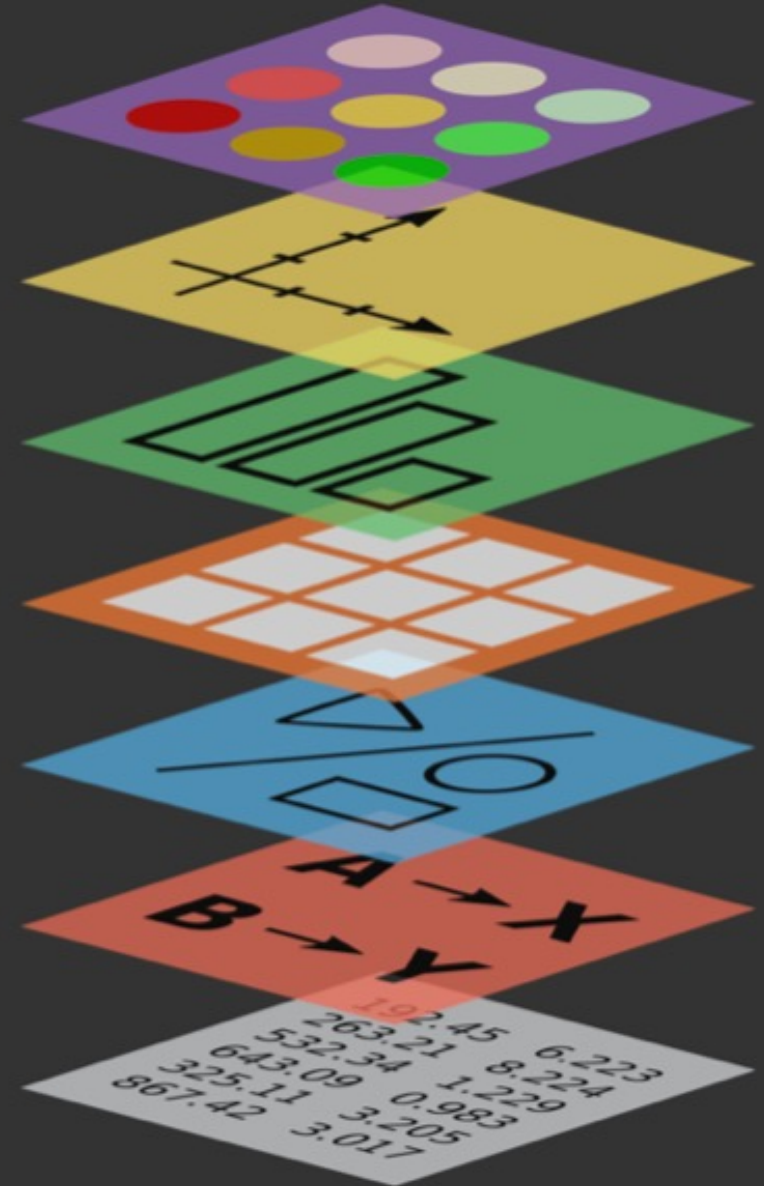
Statistics and Computing

Leland Wilkinson

The Grammar of Graphics



Theme
Coordinates
Statistics
Facets
Geometries
Aesthetics
Data



PLOTNINE EXAMPLES

Aesthetics, geoms, coordinates, scales, themes

The Titanic Data Set www.kaggle.com/c/titanic/data



Getting Started Prediction Competition

Titanic: Machine Learning from Disaster

Start here! Predict survival on the Titanic and get familiar with ML basics



Kaggle · 22,686 teams · Ongoing

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
-------------	----------	--------	------	-----	-----	-------	-------	--------	------	-------	----------

The Data

Variable	Definition	Key
survival	Survival	0 = No, 1 = Yes
pclass	Ticket class	1 = 1st, 2 = 2nd, 3 = 3rd
sex	Sex	
Age	Age in years	
sibsp	# of siblings / spouses aboard the Titanic	
parch	# of parents / children aboard the Titanic	
ticket	Ticket number	
fare	Passenger fare	
cabin	Cabin number	
embarked	Port of Embarkation	C = Cherbourg, Q = Queenstown, S = Southampton

[illegible]

How to go from data to insight?

PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
1	0	3	Braund, Mr. Owen Harris	male	22.00	1	0	A/5 21171	7.2500		S
2	1	1	Cummings, Mrs. John Bradley (Florence Briggs Thayer)	female	38.00	1	0	PC 17599	71.2833	C85	C
3	1	3	Heikkinen, Miss. Laina	female	26.00	0	0	STON/O2. 3101282	7.9250		S
4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.00	1	0	113803	53.1000	C123	S
5	0	3	Allen, Mr. William Henry	male	35.00	0	0	373450	8.0500		S
6	0	3	Moran, Mr. James	male	NA	0	0	330877	8.4583		Q
7	0	1	McCarthy, Mr. Timothy J	male	54.00	0	0	17463	51.8625	E46	S
8	0	3	Palsson, Master. Gosta Leonard	male	2.00	3	1	349909	21.0750		S
9	1	3	Johnson, Mrs. Oscar W (Elisabeth Vilhelmina Berg)	female	27.00	0	2	347742	11.1333		S
10	1	2	Nasser, Mrs. Nicholas (Adele Achem)	female	14.00	1	0	237736	30.0708		C
11	1	3	Sandstrom, Miss. Marguerite Rut	female	4.00	1	1	PP 9549	16.7000	G6	S
12	1	1	Bonnell, Miss. Elizabeth	female	58.00	0	0	113783	26.5500	C103	S
13	0	3	Saunderscock, Mr. William Henry	male	20.00	0	0	A/5. 2151	8.0500		
14	0	3	Andersson, Mr. Anders Johan	male	39.00	1	5	347082	31.2750		
15	0	3	Vestrom, Miss. Hulda Amanda Adolfina	female	14.00	0	0	350406	7.8542		
16	1	2	Hewlett, Mrs. (Mary D Kingcome)	female	55.00	0	0	248706	16.0000		
17	0	3	Rice, Master. Eugene	male	2.00	4	1	382652	29.1250		
18	1	2	Williams, Mr. Charles Eugene	male	NA	0	0	244373	13.0000		
19	0	3	Vander Planke, Mrs. Julius (Emelia Maria Vandemoorte...)	female	31.00	1	0	345763	18.0000		
20	1	3	Masselmani, Mrs. Fatima	female	NA	0	0	2649	7.2250		
21	0	2	Fynney, Mr. Joseph J	male	35.00	0	0	239865	26.0000		
22	1	2	Beesley, Mr. Lawrence	male	34.00	0	0	248698	13.0000	D56	
23	1	3	McGowan, Miss. Anna "Annie"	female	15.00	0	0	330923	8.0292		
24	1	1	Sloper, Mr. William Thompson	male	28.00	0	0	113788	35.5000	A6	
25	0	3	Palsson, Miss. Torborg Danira	female	8.00	3	1	349909	21.0750		
26	1	3	Asplund, Mrs. Carl Oscar (Selma Augusta Emilia Johan...)	female	38.00	1	5	347077	31.3875		

a statistical graphic is a **mapping** from **data** to **aesthetic** attributes (color, shape, size) of **geometric** objects (points, lines, bars)

```
ggplot( data=titanic,
        mapping=aes(x = Sex, fill=Survived)) +
  theme_bw() +
  geom_bar() +
  labs(y = "Passenger Count",
        title = "Titanic Survival Rates by Sex")
```

